

next year. The Korean giant has two research centres in India, which work for its global projects.

Samsung's story clearly indicates that electronics world has traction for the Indian talent. The nation is witnessing a talent war for engineers and technical professionals.

"Infineon India's main incentive for setting up operations in India was the country's abundance of highly skilled professionals in the areas of hardware and software development and R&D. This is now supplemented by the company's attraction to the Indian semiconductor market. According to industry analysts, India's markets for automotive and industrial electronics and smart cards are estimated to reach two billion US dollars by 2015," informs Shenoy.

According to a recent study by management consultancy Zinnov, hiring in R&D domain is expected to grow by 15 per cent this fiscal. The study suggests that operating cost of R&D centres in India has gone up by 9 per cent. Despite this, operating cost in India is still 25 per cent lower than in China.

Chandramouli C.S., management consulting director, Zinnov, said, "The increase in operating cost, beginning last year, will not dampen the investment mood. MNCs are looking at ramping up operations and continue to invest in value creation and innovation in India."

Interestingly, Tier-II locations in India are emerging fast and offer up to 40-50 per cent savings on cost. "The MNCs in India have started setting up their secondary R&D centres for non-core work in tier-II locations in India such as Madurai, Chandigarh, Baroda, Coimbatore and Bhubaneswar as these are 40 to 50 per cent more cost-effective than Tier-I locations such as Bengaluru, Pune or Chennai," according to Chandramouli.

Looking ahead, we can expect a 10 to 15 per cent growth in the next couple of years in R&D in India, with growth mostly in engineering and embedded systems. The new companies that are being established were initially into sustenance and are now focussing more on innova-

Major contributors to this report



Guruswamy Ganesh
Vice president and country manager, Freescale Semiconductor India

Indranil Goswami
Head-Lighting Application Services, Philips Lighting India



Jitendra Chaddah
Director, Intel India Strategic Development & Operations

Dr Mano Manoharan
General manager, operations, GE Global Research, and technology leader, manufacturing and materials, John F. Welch Technology Center



Vinay Shenoy
Managing director, Infineon Technology India

tion and leadership, which is completely different from what they actually do.

Of course, there are some roadblocks too that India needs to overcome in order to enjoy a smooth ride as an R&D hub.

Companies like Samsung, Intel and AMD are facing talent crunch in their R&D centres in India, making them turn towards Indian universities for talent.

"There is surely a talent crunch when it comes to candidates with PhDs. One of the other challenges that we face is that as a technology company we need to build the depth in terms of engineers' skillsets. Since there is a strong service mindset in people, after working for a year or so sometimes they start feeling nervous doing the same thing and express the desire to change jobs. That mindset needs to change, as in technology and R&D depth

is very critical and important," says Jitendra Chaddah, director, Intel India Strategic Development & Operations.

Intel's rival, the California based \$6.49-billion AMD Inc., is also facing a talent crunch in its design centres in Hyderabad and Bengaluru.

Kiranmai Pendyala, head-HR, AMD India, says, "We are acquiring senior professionals from various markets like Bengaluru, Delhi, Hyderabad, Pune, Singapore and the US. Attrition has increased in 2010-11 as the market is ripe and growth opportunities exist in the ecosystem. Hiring cost has definitely increased, given the market dynamics."

Lack of adequate physical infrastructure (lack of regular power supply, poor roads) definitely poses a challenge to operate smoothly in India and also in faster growth of already existing units.

"A lot needs to be done in the area of industry and academia interface to ensure that the technologies that students learn in the class are the latest that the industry is working on. While this has already started to a great extent, penetration needs to take place across the universities in India. Also, the R&Ds happening in the country are working towards providing solutions for the global market while the demand for unique products to serve India's growing market is increasing. It is important that R&D players invest for home-grown solutions as well, which can range from solar energy for generating power to cost-effective connectivity solutions," says Guruswamy.

"To enhance the R&D eco-system, engineering institutes should emphasise postgraduate education in technologies relevant for the Indian market; for example, energy efficiency, automotive safety and security. This will enable more relevant collaboration between the industry and academia," suggests Shenoy.

"Most important of all is the support from the government to leverage this strength and continue to provide all help in encouraging more growth by giving tax incentives to help companies grow their R&D labs here, which benefits India immensely," adds Guruswamy.